

Week 5, Day 1 — student handout

Umesh Indrajith · First SMS HTTP API. No DB / JWT today.

OUTCOMES

- HTTP method, path, status, JSON body.
- Tiny net/http server, then Gin.
- GET /health and GET /api/hello on :8080.
- Prove with curl and Postman.

ADD GIN

```
cd backend
go get github.com/gin-gonic/gin
go mod tidy
```

HEALTH (GIN)

```
r := gin.Default()
r.GET("/health", func(c *gin.Context) {
    c.JSON(http.StatusOK, gin.H{
        "status": "ok",
        "message": "Backend is running",
    })
})
r.GET("/api/hello", func(c *gin.Context) {
    c.JSON(http.StatusOK, gin.H{"message": "hello from SMS"})
})
r.Run(":8080")
```

PROVE IT

```
go run ./cmd/server
curl -i http://localhost:8080/health
curl -s http://localhost:8080/api/hello
```

STATUS CHEAT SHEET

Code	Meaning
200	OK
400	Bad request
404	Not found
500	Server error

IF IT BREAKS

- refused → start the server
- port in use → kill old process
- gin missing → go get in backend/
- Postman https → use http://

HOMework

PR with health + hello + README curl + Postman 200 proof. Stretch: hard-coded GET /api/students array. Details: homework.md.